





Early–late asymmetry of the muon density with the UMD

Selection bias, core reconstruction, and how to recover it

Lautaro Silva Pizzi

with J. M. Figueira and F. Sánchez

ITeDA (CNEA-CONICET-UNSAM) — FCEN-UBA

UMD Foundations Meeting — 4 November 2026

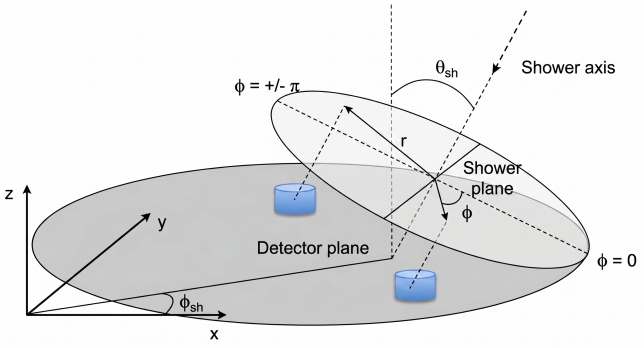
This talk is about the azimuthal asymmetry of the muon density, measured underground with the UMD. I'll show that the asymmetry is clearly there in simulations, then two analysis effects that can hide it or even flip its sign, and finally how we can account for one of them.

Transition: Let me first say what we measure.

The observable: first harmonic in the shower plane

- ϕ: station azimuth in the **shower plane**.
- Vertical shower: circular symmetry.
- Inclined shower: two distinct regions.
  - **early** ($\phi \approx 0°$): reached first, through *less* atmosphere.
  - **late** ($\phi \approx \pm 180°$): reached later, through *more* atmosphere.
- $A_1 > 0$: early excess; $A_1 < 0$: late excess.
- Fitted on an **ensemble** of showers, per (r, θ) bin.

$$\rho(r, \phi) = \langle \rho(r) \rangle \, (1 + A_1 \cos \phi)$$



The asymmetry of the total SD signal in inclined showers is well known. The question here is what happens with muons alone, which is what the UMD gives us: shielded, so free of the electromagnetic component. The observable is the first-harmonic amplitude A1 of the muon density in the shower plane. For a vertical shower the density is circularly symmetric. When the shower is inclined, two regions appear: the early side, reached first through less atmosphere, and the late side, reached later through more atmosphere. A positive A1 means more muons on the early side. A1 is fitted on an ensemble of showers, not event by event.

Transition: We start in the cleanest setup we have.

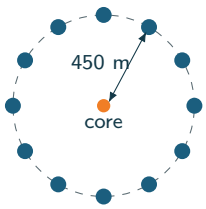
The reference: Monte Carlo geometry

Simulations and the Dense Ring

- Official CORSIKA + *Offline* production, with SD-750 and UMD:

Hadronic models	SIBYLL 2.3e (main), QGSJET-III.01, EPOS-LHC-R
Primaries	p, He, O, Fe
Energy	$10^{17.5}$–10^{18} eV
Zenith angle	$\theta \leq 65^\circ$

- **Dense Ring**: 12 virtual stations at $r = 450$ m from the **true** axis, every 30° in the shower plane. Same r for all: the radial fall-off of the density cannot produce a modulation in ϕ.

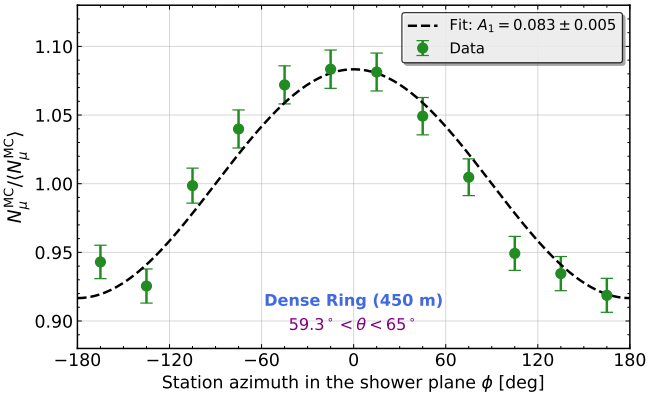


Shower plane.

These are the simulations: the official collaboration production, CORSIKA plus Offline, with the 750 meter array and the UMD. Most of what I show is protons with SIBYLL 2.3e between ten to the 17.5 and ten to the 18 electronvolts. To isolate the effect we start with a synthetic setup, the Dense Ring: twelve virtual stations placed in the simulation at 450 meters from the true shower axis, every thirty degrees. Since all stations are at the same distance, the steep fall of the density with distance cannot produce a modulation in azimuth.

Transition: This is what the muon density looks like on the ring.

A clear modulation on the Dense Ring

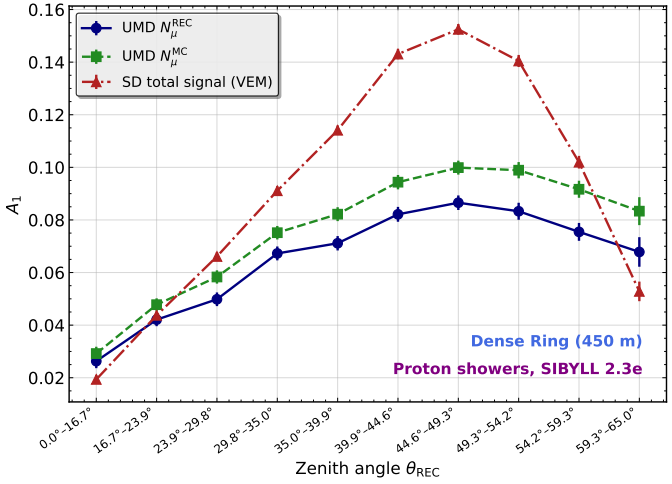


- Injected muon number N_μ^{MC}, $59.3^\circ < \theta < 65^\circ$.
- Maximum in the **early** region ($\phi \approx 0$).
- Early–late difference $\approx 17\%$ peak to peak ($\pm 8\%$ around the mean).

Here are the twelve ring stations for the most inclined bin, using the injected muon numbers. The density peaks on the early side and the first harmonic gives an amplitude of 0.083: the early side has about eight percent more muons than the mean, the late side eight percent fewer. With the reconstructed muon numbers the same bin gives 0.068; that difference is the counting reconstruction, in the backup. (If asked: the Dense Ring zenith bins are in reconstructed zenith angle.)

Transition: How does it change with zenith angle?

Comparison: SD vs. UMD on the Dense Ring



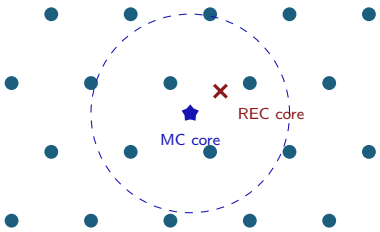
- UMD: $A_1 > 0$, grows with θ up to $\sim$ 45–50°.
- UMD: lower amplitude with reconstructed (N_μ^{REC}) than with injected (N_μ^{MC}) muons, same shape.
- SD total signal: larger than the UMD at $\sim$ 20–60° (EM-dominated); drops sharply above $\sim$ 50°.
- SD below the UMD only at $\theta \gtrsim 60°$.

The muon asymmetry is positive and grows with inclination up to about forty-five to fifty degrees. The SD total signal is larger over most of the range, because it is dominated by the electromagnetic component, which is strongly attenuated after the shower maximum. Above about fifty degrees the electromagnetic part is mostly absorbed and the SD asymmetry drops quickly; only in the last bin, above about sixty degrees, does it fall below the UMD. Note also the gap between the two UMD curves: the reconstructed muon numbers give a lower amplitude than the injected ones, with the same shape. That comes from the muon-counting reconstruction; it is in the backup.

Transition: The ring gives us one radius. For the radial dependence we need the real Infill array.

The Infill array: two geometries

- Now the real Infill stations (750 m): the distance to the axis varies from station to station $\Rightarrow$ we can study $A_1(r)$.
- r and ϕ of each station depend on where the shower axis is. Two geometries:
  - **MC**: true core and direction;
  - **REC**: reconstructed core and direction.
- The main difference between them is the **core position**.

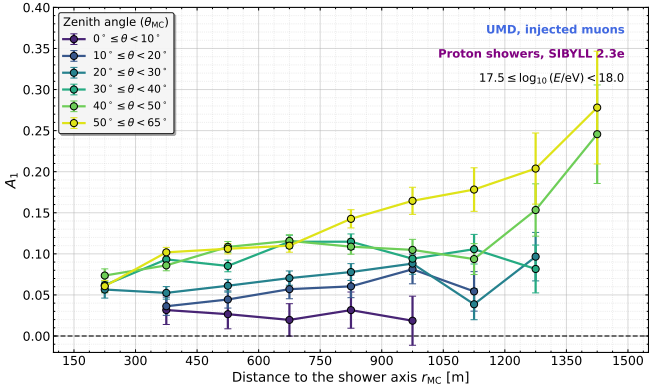


Stations every 750 m. With another core, r and ϕ of every station change.

Still simulations, but now with the real Infill stations. Their distances to the axis vary, so we can follow A1 as a function of r. The distance and azimuth of each station depend on where the shower axis is, and there are two choices: the true core and direction from the simulation, or the reconstructed ones. The main difference between the two is the core position. We start with the true one.

Transition: This is the UMD asymmetry with the true geometry.

UMD radial dependence on the Infill (MC geometry)



- **UMD only** (injected muons); r, ϕ from the **true** core and direction.
- Radial bins of $\Delta r = 150$ m; zenith bins of 10°.
- $\theta < 10^\circ$: compatible with zero.
- 10–40°: grows with r, then **flattens**.
- $\theta \geq 40^\circ$: **grows with** r.

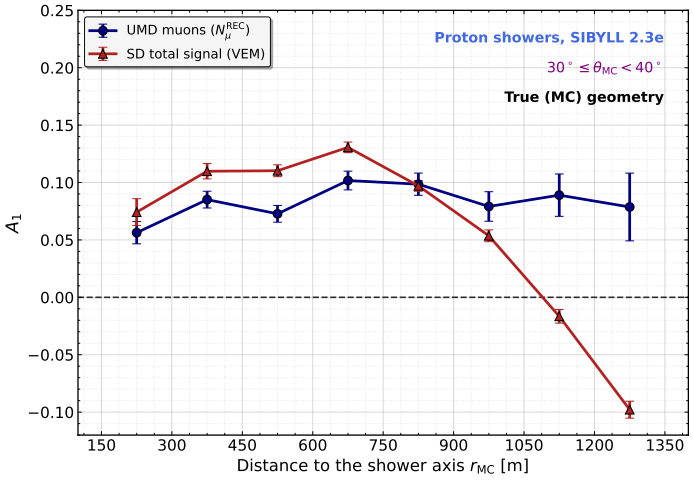
This is our reference.

This is the UMD alone, with the injected muon numbers and the true core and direction, in radial bins of 150 meters and zenith bins of ten degrees. Near-vertical showers show no significant asymmetry. Between ten and forty degrees it grows with distance and then flattens. Above forty degrees it grows with distance. (If asked: at forty to fifty degrees it stays around 0.1 and rises only beyond about twelve hundred meters, where the statistics are lower.) This is what an ideal analysis should find.

Transition: Now we compare with the surface detector in the same showers.

The SD inversion and the selection bias

The SD comparison: inversion far from the core

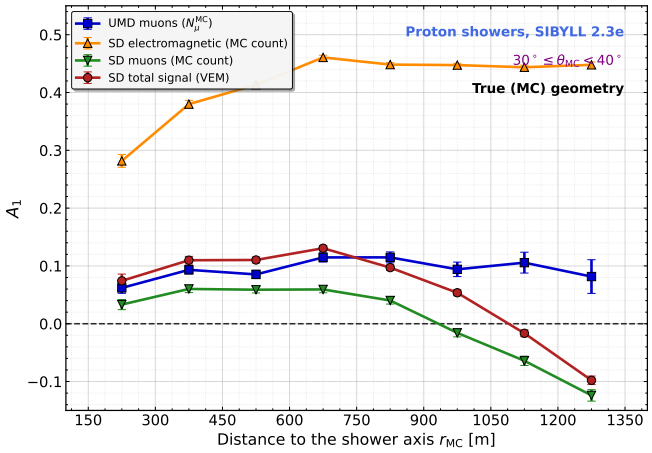


- We now compare the UMD and SD asymmetries in the same Infill showers.
- MC geometry, $30^\circ \leq \theta < 40^\circ$.
- UMD: positive at all r.
- SD total: a maximum, then negative far from the core.

Now we compare the UMD with the surface detector, on the same Infill showers and with the same true geometry. The UMD stays positive at all distances. The total SD signal peaks, then decreases, and far from the core becomes clearly negative, an excess on the late side. Since the core and direction are the true ones, this cannot be a reconstruction error.

Transition: Which part of the tank signal is responsible?

It is the SD muon component that inverts



- MC particle counts in the tank, split by component.
- **EM**: positive at all r.
- **SD muons**: positive near the core, negative beyond $\sim$ 900 m.
- The SD total follows its muons at large r.

Same quantity, opposite sign in UMD and SD.

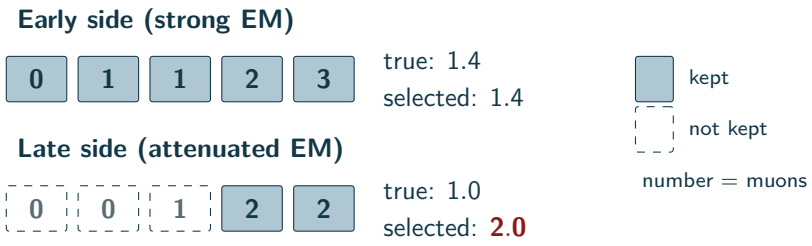
Splitting the tank signal with the simulation truth, the electromagnetic part is positive everywhere. The muon part is what inverts, and the total SD signal follows it far from the core. So we compare the same physical quantity, muons crossing a detector, and get opposite signs in the UMD and in the SD. Our analytic model of muon propagation does not produce a negative asymmetry at these distances, so we looked at how the observable is built.

Transition: The key point: we never average over all stations.

Hypothesis: the station selection biases the average

HYPOTHESIS

A station enters the analysis only if it **belongs to the reconstructed event** (`HasStation`). If the probability that a station is kept depends on its own muon content, and this dependence is not the same on the early and late sides, the **regional averages are biased**.



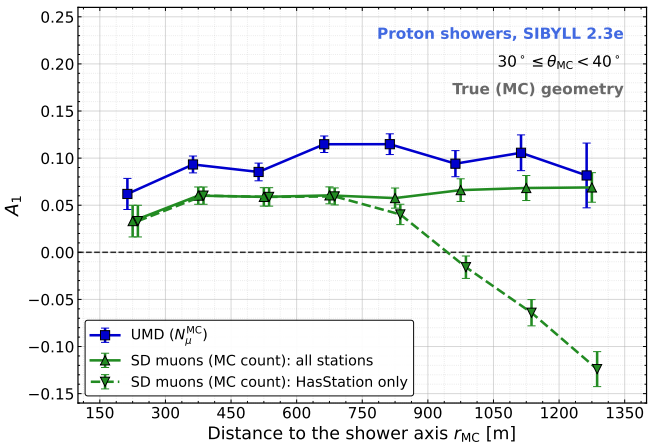
The sign can flip **without moving a single muon**.

Illustrative cartoon, not simulation output.

Every curve so far is an average over stations that are part of the reconstructed event. For the UMD that seemed natural, since it has no trigger of its own. But suppose the chance that a station is kept depends on how many muons it has. On the early side the electromagnetic signal is strong enough that almost every station is kept. On the late side it is more attenuated, so mostly the stations with more muons survive. The late average is pulled up, and the asymmetry can change sign although no muon was created or moved. This is a hypothesis; the next two slides test it.

Transition: The direct test is to remove the requirement.

Test 1: without HasStation, the inversion disappears



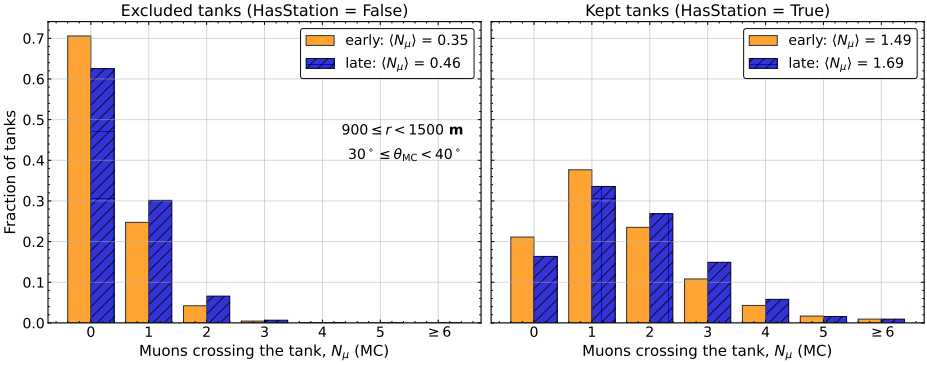
- Same MC counts, coordinates and fit: only **which stations** enter changes.
- All stations: SD muons **positive at all** r ($+0.03$ to $+0.07$).
- Same sign and trend as the UMD, **but not the same values**.

Same Monte Carlo muon counts in the tanks, same coordinates, same fit; only which stations enter changes. Over all simulated stations the SD muon asymmetry is positive at all distances, between plus three and plus seven percent. Requiring the station to be in the reconstructed event changes nothing up to about 750 meters, where practically every station is kept, and then drives the asymmetry down to minus 0.12 at twelve to thirteen hundred meters: plus 0.069 plus or minus 0.016 over all stations, against minus 0.124 plus or minus 0.019 with the requirement. The errors come from a bootstrap over the 960 showers, keeping the stations of each shower together. So the sign flip is a property of the station sample. Without the selection the SD muons have the same sign and trend as the UMD, but between 150 and 900 meters the UMD is still 0.03 to 0.06 higher; detector response and threshold are candidates, this is open.

Transition: If this is a trigger effect, there is a direct check.

Test 2: the excluded tanks should hold very few muons

- If the inversion comes from the **trigger**, the tanks dropped by the reconstruction (`HasStation` = False) should have been crossed, on average, by at most one muon, too few for the station to trigger.
- Excluded tanks: $\leq 1\,\mu$ in 95% (early) and 93% (late) of the cases.



If the inversion is a trigger effect, there is a clear prediction: the tanks that the reconstruction drops should have been crossed by at most about one muon. If we found excluded tanks with several muons, a trigger explanation would not be credible. So we count the Monte Carlo muons in the excluded tanks. Ninety-five percent of the excluded early tanks and ninety-three percent of the late ones have at most one muon. The excluded late tanks have about half the electromagnetic particles of the early ones, 14 against 28, so on the late side we lose tanks that on the early side would have been kept thanks to their electromagnetic signal. That is why, among the kept tanks, the late ones carry more muons than the early ones, even though the true density is higher on the early side, and why only eighteen percent of the late stations are kept far from the core, against forty-two percent of the early ones. This supports the trigger picture, but HasStation also includes readout and reconstruction, and we have not separated them, so it is consistent with a trigger effect, not a proof. (Early: $|\phi| < 60°$, late: $|\phi| > 120°$; MC geometry.)

Transition: So far, true core and direction. In real data we only have the reconstructed ones.

Reconstructed geometry and the core position

With REC geometry, the UMD asymmetry vanishes

- Same UMD modules, same **injected** muon number, $450 \leq r < 600$ m.

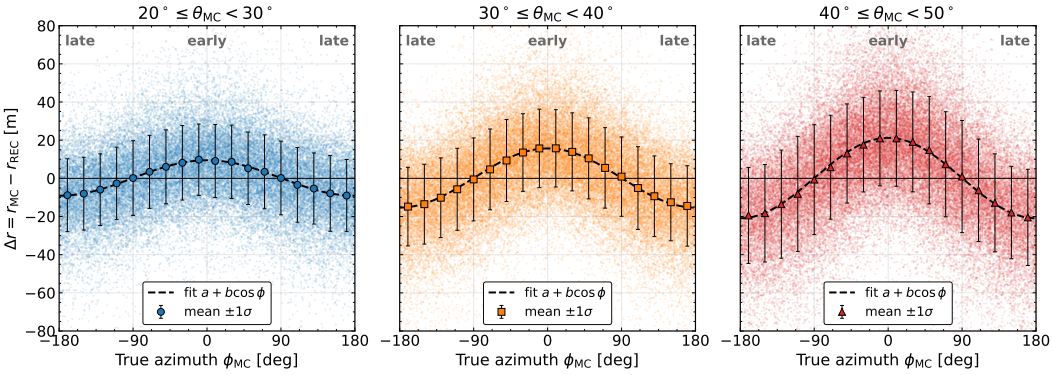
θ	A_1 Monte Carlo	A_1 Reconstructed
20°–30°	0.062	0.011
30°–40°	0.086	0.002
40°–50°	0.108	−0.002

- With the **reconstructed core**, the asymmetry is lost almost completely.
- The effect comes from the **offset between the true and the reconstructed core positions**.

Same UMD modules, same injected muon numbers; we only change the geometry used to compute each station's distance and azimuth. With the reconstructed core the asymmetry essentially disappears: at forty to fifty degrees it goes from 0.108 to zero. We also swapped the two quantities one at a time: using the reconstructed or the true azimuth gives the same A1 within 0.002, so everything comes from the distance to the reconstructed core. (If asked: farther out, at forty to fifty degrees and 600 to 1050 meters, the asymmetry survives but reduced, below 0.05 against about 0.11.)

Transition: So where is the reconstructed core?

The reconstructed core is shifted toward the early region



- $\Delta r = r_{\mathrm{MC}} - r_{\mathrm{REC}}$ vs. ϕ can be fitted by $a + b\cos\phi$: early stations are placed **closer** to the core, late ones **farther**, with $b \simeq 9$, 15, 21 m, increasing with θ.
- Likely cause: the reconstruction LDF is **symmetric in** ϕ; it absorbs the early excess by **moving the core toward the early side, where there is more signal**.

For each station we compare its distance to the true and to the reconstructed core, as a function of its true azimuth. If the core error were random, this would be flat. Instead it follows a cosine: early stations are placed closer to the core than they are, late stations farther. That is the signature of a systematic shift of the reconstructed core toward the early side, by about nine, fifteen and twenty-one meters in these three zenith bins. A likely reason: the LDF used in the reconstruction is symmetric in azimuth. Faced with more signal on the early side, the only thing the fit can do is move the core toward it. (Infill stations with reconstructed distance below 1400 meters.)

Transition: What does a shift of this size do to the LDF, and to A1?

A shifted core adds a $\cos\phi$ term to the LDF

- Approximating the LDF locally as $\rho \propto r^{-\beta}$ and correcting it for the core shift with an azimuthal dependence, $r \to r + b\cos\phi$:

$$\rho(r + b\cos\phi) \simeq \rho(r)\left[1 - \beta\,\frac{b}{r}\cos\phi\right] \;\Rightarrow\; \boxed{\Delta A_1 \simeq -\beta\,\frac{b}{r}}$$

θ	A_1 (MC)	A_1 (REC)	ΔA_1	$A_1^{\mathrm{REC}} + \beta b/r$
20–30°	0.062	0.011	−0.051	**0.062**
30–40°	0.086	0.002	−0.084	**0.083**
40–50°	0.108	−0.002	−0.110	**0.100**

- The shift accounts for **93–100%** of the loss; adding it back **recovers the MC reference asymmetry** to $\lesssim 0.01$.

Locally the muon LDF falls as a power law with slope beta, between about 2.4 and 2.9 at these distances. If the reconstructed core is shifted by b toward the early side, a station we place at distance r on the early side is really at r plus b, so it has fewer muons than we think, and a late station has more. To first order this multiplies the LDF by one minus beta b over r times cos phi: a cosine modulation with the opposite sign to the physical one. So the reconstructed asymmetry should be lower by beta b over r. The prediction, minus 0.051, 0.081 and 0.102, matches the observed loss in the three zenith bins. Read the other way: adding beta b over r back to the reconstructed value recovers the Monte Carlo reference within about 0.01. The exact agreement depends on the beta fit range and on the effective r of the bin at the ten percent level.

Transition: So what would it take to apply this to real data?

Recovering the asymmetry: what it needs, what is missing

What the correction needs

- $b(\theta)$: mean core shift, early–late direction. Measured with MC truth.
- β: local slope of the muon LDF. From MC here; in principle measurable in data.
- Apply: move the core back by b, or add $\beta b/r$ to A_1.

Established

The loss is **systematic**, not resolution, and predicted by $\beta b/r$ (450–600 m, 20–50°).

Not done yet

- Move the core and recompute $A_1(r)$ at all r.
- Counting bias (N_μ^{REC}, 5–23%): separate effect.
- **For data** (no true core), b must come from:
  - $b(\theta)$ parametrized on simulations and **validated on unseen ones** (other showers, primaries, models; tested so far: p, SIBYLL 2.3e);
  - or an LDF fit with an azimuthal term in the reconstruction.

The correction has two ingredients. The slope beta is a property of the lateral distribution and could in principle be measured from the data themselves. The shift b is the delicate part: so far we measure it against the true core, which only exists in simulations. In data there is no true core, so there are two ways. One is to parametrize b as a function of zenith angle on one set of simulations and check, as in a train and test split, that it recovers the asymmetry on simulations it has not seen: other showers, other primaries, other hadronic models. That tells us how model- and composition-dependent the correction is; it does not test the differences between data and simulations, such as the muon deficit. The other way is to remove the shift at its origin, with a reconstruction LDF that allows an azimuthal modulation. At Infill energies the station multiplicity makes that hard event by event, but it is worth trying for the most energetic events. The test is done in one radial bin, for protons with one hadronic model, and with injected muon numbers. The counting bias of the UMD reconstruction is a separate, smaller effect.

Transition: Let me summarize.

Conclusions

Conclusions

1. In simulations the UMD muon density has a **positive early–late asymmetry**: it grows with r for inclined showers and is roughly flat at intermediate angles.
2. The SD muon asymmetry inverts far from the core **only** when averaging over the stations kept by the reconstruction: the selection acts **station by station**, not shower by shower. Over all stations it is positive. The excluded tanks hold **mostly** ≤ 1 muon: **consistent with** a trigger effect, not yet proven.
3. With the reconstructed geometry the UMD asymmetry nearly vanishes near the core: the reconstructed core is shifted toward the early region by $b \approx 9$–21 m.
4. This loss follows $\Delta A_1 \simeq -\beta\, b/r$; adding it back recovers the MC reference. Applying it to data requires b **without MC truth**.

Data–MC comparisons need the same station selection and the same geometry.

Four points. The muon asymmetry is there, and it behaves as expected with zenith angle and distance. What we measure depends on how we build the observable: which stations we average over can flip the sign, and the reconstructed core can erase it. The second effect is simple and predictable, so it can be accounted for, and any comparison between data and simulation must use the same station selection and geometry. The open step is to determine the core shift without Monte Carlo truth.

Next steps

- Parametrize the correction parameters vs. zenith angle: $b(\theta)$ and $\beta(\theta)$, at fixed energy ($17.5 \leq \lg E < 18.0$), and validate the recovery on independent simulations.
- Apply the full correction (core shift + LDF term) over the whole radial range and compare with the MC reference.
- Dependence on primary and hadronic model.
- Separate trigger, readout and reconstruction inside `HasStation`.
- Apply to field data.

The natural next step is to parametrize b and beta as functions of zenith angle, so that the correction can be applied without a per-bin Monte Carlo comparison. Our sample covers a single energy range, so this is a zenith-angle parametrization only, not an energy one.

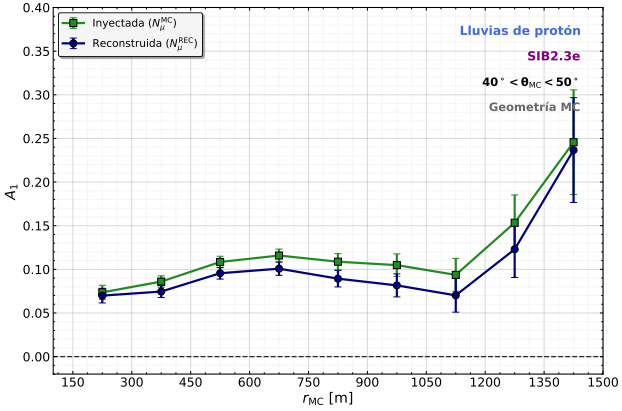


Thank you!

Questions?

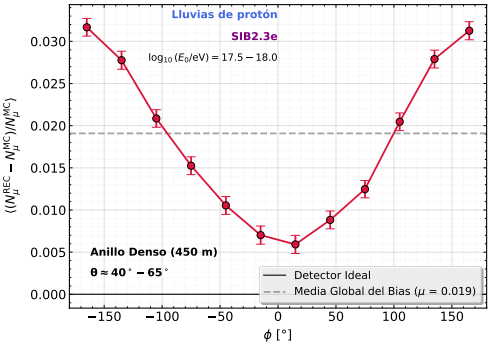
Image: A. Chantelauze, S. Staffi, L. Bret (Pierre Auger Observatory)

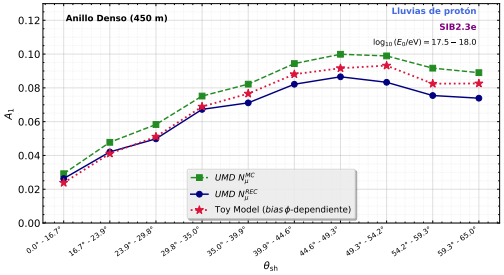
Backup: muon-counting reconstruction (MC geometry)



- 40°–50°, true core and direction.
- Reconstructed muon number lowers A_1 by 5–23%, same shape.
- A separate effect from the core shift.

Backup: the counting bias depends on azimuth

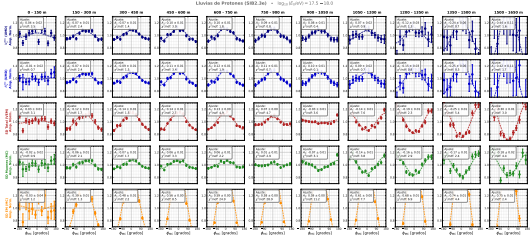




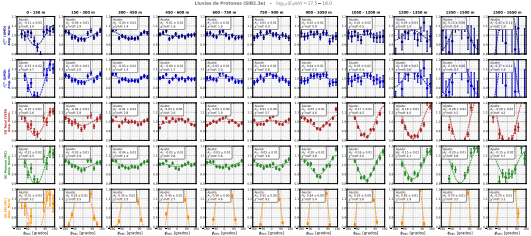
- Mean relative residual $(N^{\mathrm{REC}} - N^{\mathrm{MC}})/N^{\mathrm{MC}}$ is larger on the late side (fewer muons) $\Rightarrow$ lower A_1.
- Bootstrap with the per-region residuals reproduces A_1^{REC} up to $\theta \approx 35°$; above, half to two thirds of the loss. Remnant open.

Backup: fit grids, MC and REC geometry (40°–50°)

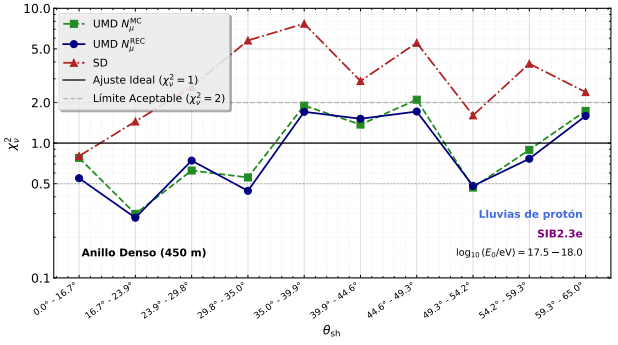
MC geometry



REC geometry



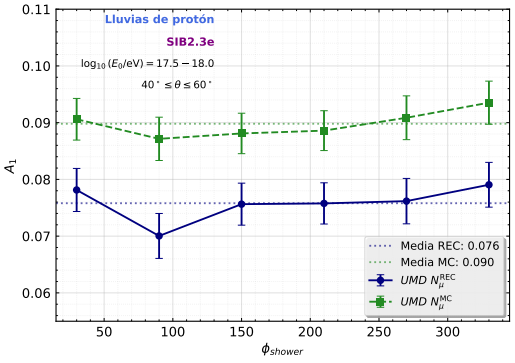
Backup: fit quality on the Dense Ring



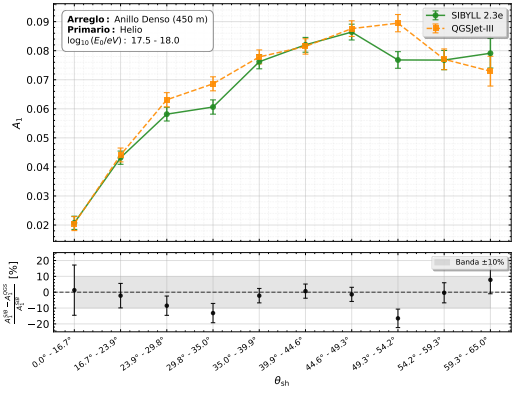
- UMD, MC and REC: $\chi^2_\nu \approx 1$.
- SD: χ^2_ν up to ~ 10; with such small errors the fit is sensitive to higher harmonics and to correlations between stations of one shower.

Backup: robustness

SIMULATIONS – DENSE RING

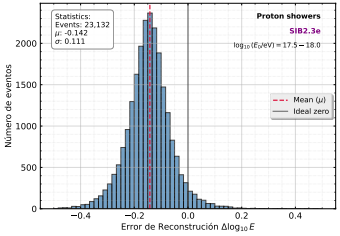


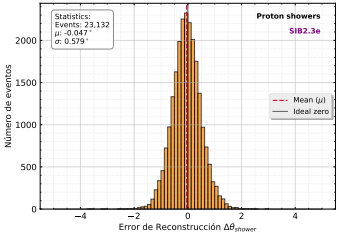
Independent of the shower arrival azimuth (40–60°).

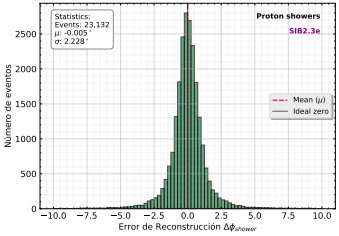


Helium: SIBYLL 2.3e vs. QGSJET-III within ±10% over most of the range.

Backup: reconstruction performance







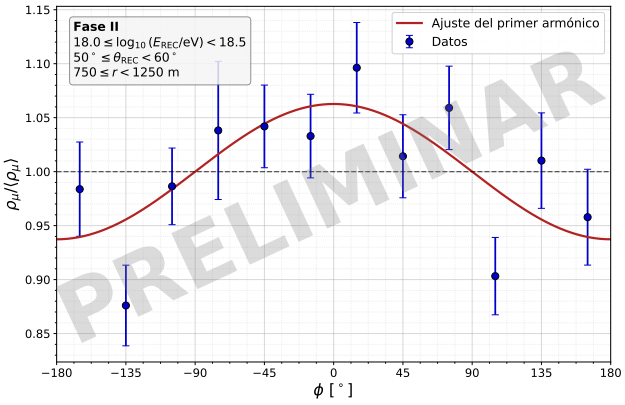
- Energy: $\mu \approx -0.142$, $\sigma \approx 0.111$ in $\Delta \log_{10} E$.
- θ: bias $-0.05°$, $\sigma = 0.58°$. ϕ: bias $-0.005°$, $\sigma = 2.2°$.

Backup: why the core shift does not affect the Dense Ring

- The ring stations are placed during the simulation at 450 m from the **true** axis: the shifted core does not make the ring eccentric.
- It only rotates the azimuth assigned to each station, by $\sim b/r \approx 2^\circ$, much less than the 30° spacing.
- It affects N_μ^{MC} and N_μ^{REC} in the same way, so it cannot explain the MC–REC gap on the ring (that gap is the counting bias).

Backup: field data (preliminary)

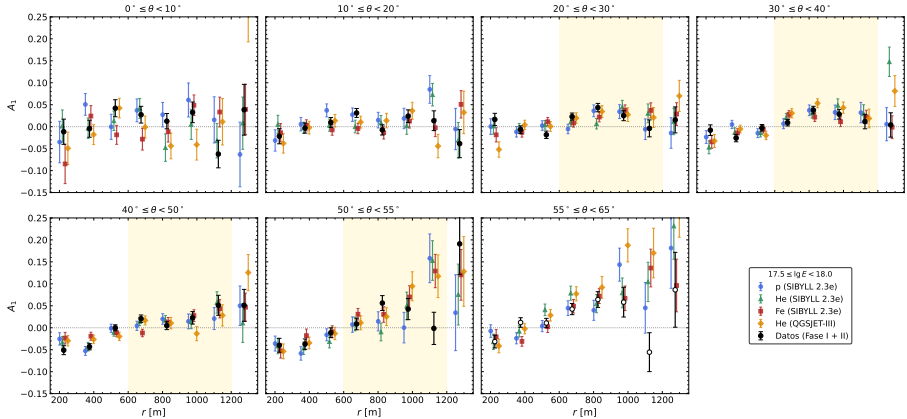
FIELD DATA – PRELIMINARY



- UMD muon density vs. ϕ, normalized to its mean; Phase II, cuts in the box.
- Maximum at $\phi \approx 0$: **early excess**, as in the simulations.
- **Preliminary**: no core-shift correction, no systematics, no MC above 10^{18} eV for this bin.

Backup: $A_1(r)$ in field data vs. simulations

FIELD DATA – PRELIMINARY



UMD Infill, Phase I + Phase II, 6T5, $17.5 \leq \lg(E/\text{eV}) < 18.0$, REC geometry in data and simulations.